

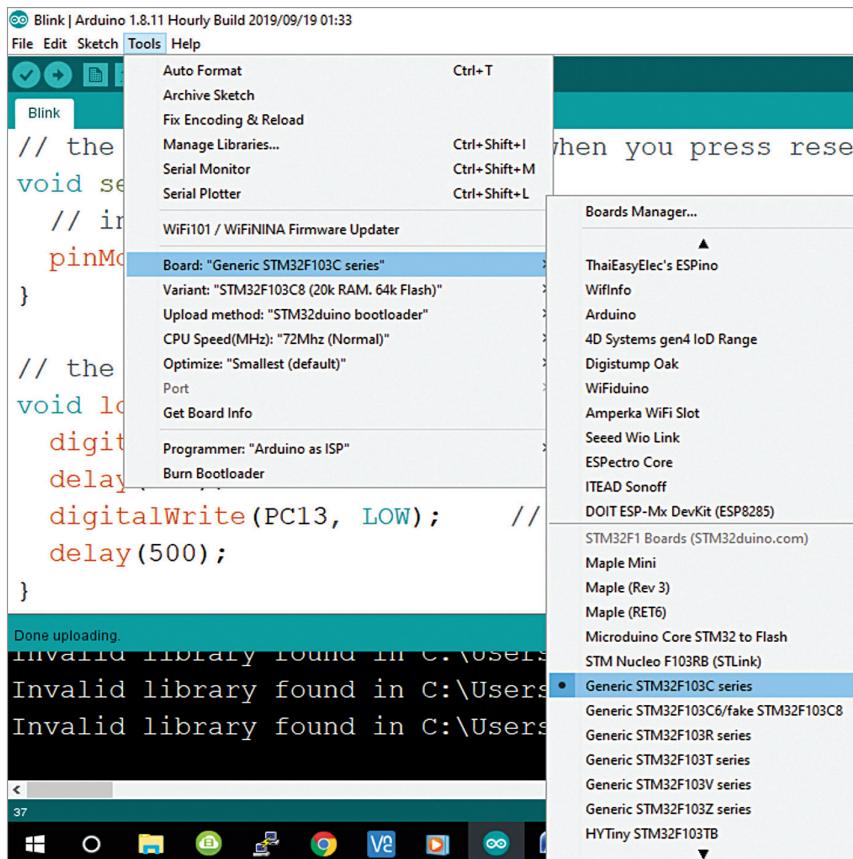


Fig. 11: STM32F103C board selection process

lation is done, go to Tools and scroll down to find the Generic STM32F103C series, as shown in Fig. 11. Make sure the variant is 64k Flash type, CPU speed is 72MHz; change the upload method to Serial.

Step 6. Connect the Micro USB board to the computer and check to which COM port the STM32 board is connected, using device manager. Select the same port number in Tools→Port.

Step 7. After all the changes are made, check the bottom right corner of the Arduino IDE. You will notice the generic STM32F103C (Blue Pill) board is connected to COM9, as shown in Fig. 12. Now the Arduino IDE is ready to program the STM32 Blue Pill development board.

To operate the circuit the software program needs to be loaded into the internal memory of STM32. The program is written in Arduino programming language Sketch. Ar-

duino IDE 1.8.11 is used to compile and upload the program. The project needs following external header file for programming:

```
#include <MD_MAX72xx.h> :
```

The library implements functions that allow the MAX72xx to be used for LED matrices (64 individual LEDs), allowing the programmer to use the LED matrix as a pixel device, displaying graphics elements much like any other pixel addressable display.

Construction and testing

After uploading the source code *mains.ino* into STM32, complete wiring of the main circuit and connect 5V regulated power supply to it. After powering on, the circuit is ready to use.

EFY Note

The source code of this project is available for free download at source.efymag.com



Fig. 12: STM32F103C connection

PARTS LIST FOR 5V RPS

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- D1, D2 - 1N4007 rectifier diode

Capacitors:

- C1 - 1000µF, 25V electrolytic
- C2 - 100µF, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- X1 - 230V AC primary to 12-0-12V, 500mA secondary transformer

For testing the project, upload the code *mains.ino* (written in Arduino IDE) to the STM32 microcontroller. Since the microcontroller does not have an Arduino bootloader pre-installed, we have to use an external device—such as the FTDI module or the USB-to-TTL converter with CP 2102 driver—to transfer the code by connecting one end of the module/converter to our laptop and the other to UART pins of the microcontroller.

To connect FTDI module, connect its VCC pin to 3.3V of the STM32 module, the GND pin to the GND pin of the STM32 module, and the RX and TX pins to the PA-9 and PA-10 pins of the STM-32 module, respectively.

An important step before transferring the code to the STM32 module is to set its boot mode to Boot-1. After this you can transfer the code serially from your laptop to the STM32 module.

For testing the project, you can use a normal adaptor having a micro USB for power supply instead of the 5V regulated power supply mentioned earlier. The mobile app can then be downloaded to connect to the HC-05 module. Now data can be input to check whether the project works as expected. **EFY**



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